

Yash Joshi

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EDUCATION

PURDUE UNIVERSITY

B.S. in Mechanical Engineering

West Lafayette, IN

Expected Graduation: May 2029

- **GPA:** 3.87/4.00
- **Relevant Coursework:** Linear Algebra, Ordinary Differential Equations, Geometric and Annotation Modeling, Basic Mechanics I, EE Fundamentals I, Vertically Integrated Projects (VIP)

NEUQUA VALLEY HIGH SCHOOL

High School Diploma, Magna Cum Laude

Naperville, IL

August 2021 – May 2025

EXPERIENCE

Purdue Lunabotics

August 2025 - Present

Drivetrain Sub-team Member

- Utilized Solidworks CAD to design gearboxes and essential frame elements optimized for performance and traversal efficiency
- Reduced assembly time of gearbox systems by 20% by integrating Design for Manufacturing (DFM) principles and validating ideas through rapid prototyping
- Developed CAM within Autodesk Fusion for high precision 3-axis CNC milling of aluminum gearbox parts and waterjet cutting of plexiglass skid plate parts, with consideration of 5+ kg weight reduction goal
- Maintained detailed documentation and trade studies to be presented in multiple design reviews in an effort to inform and learn from industry partners and club members

ASME (American Society of Mechanical Engineers)

August 2025 - Present

Industrial Relations Representative

- Communicated with industry partners including Tesla and Hoosier Tire in an effort to gather meaningful sponsorships and vital organization funding
- Worked with cross-functional executive board branches to put together information sessions and recruiting events for 300+ organization members

Racing Team Member

- Utilized Autodesk Fusion and MotorXP to design an experimental YASA Axial Flux Motor with a 3000 RPM running speed
- Conducted in-depth structural analysis to ensure the efficiency and stability of the motor and mounting solution when running
- Manufactured motor components and assembly devices using 3D printing (FDM and Resin) and intentful material selection considering temperature and conductance constraints

TECHNICAL PROJECTS

Custom Electric Skateboard

- Engineered a custom electric skateboard deck in CAD, integrating a 3D-printed shell with a load-bearing steel skeleton for an optimized strength-to-weight ratio
- Conducted multi-body FEA and beam calculations to evaluate dynamic deflection and compare maximum stresses with yield strength
- Applied DFM principles to account for the tolerances of 3D printing materials and ensure efficiency in the fabrication process
- Designed a structural honeycomb pattern to decrease mass by 35% while preserving stiffness and ensuring even load distribution

Obstacle Detecting Miniature Car

- Developed an autonomous miniature car using a 3D printed chassis and Ultrasonic sensors driven by an Arduino
- Created a C++ program within the Arduino IDE to detect and avoid obstacles with 90% accuracy in testing

TECHNICAL SKILLS

CAX/Analysis Tools: Siemens NX, Solidworks, Autodesk Fusion, ANSYS, MotorXP, Aras Innovator

Technical Proficiency: GD&T, FDM 3D Printing, Arduino, Microsoft Office Suite

Languages: Python, C/C++, MATLAB, English, Spanish, Telugu